

Homework 3 Review

STAT 220B – Discussion

What HW3 was about

HW3 was the minimax homework. The four problems are four different routes to a minimax rule, ordered from the most concrete to the most abstract:

- ▶ **Problem 1:** minimax by **risk-set geometry**. A finite game, solved by drawing the risk set in $\mathbb{R}^2$ and reading off the minimax point.
- ▶ **Problem 2:** minimax by the **equalizer-Bayes** theorem. Find a Bayes rule with constant risk; constant risk plus Bayes optimality forces minimaxity.
- ▶ **Problem 3:** minimax by a **sequence of priors**. The candidate rule has constant risk equal to the limit of Bayes risks; no single prior works, but a sequence does.
- ▶ **Problem 4:** minimax on a **restricted parameter space**, via a two-point least-favorable prior.

The common engine is always the same inequality:

$$\sup_{\theta} R(\theta, \delta^*) \leq r(\pi, \delta^*) \leq r(\pi, \delta) \leq \sup_{\theta} R(\theta, \delta),$$

which says a Bayes rule with risk matching its own Bayes risk is minimax. Problems 1, 2, 4 use it directly; Problem 3 uses its limiting version.

Problem 1: Risk-set geometry

The question

Two-person zero-sum game, $\Theta = \{\theta_1, \theta_2\}$, $\mathcal{A} = \{a_1, a_2, a_3\}$, loss matrix

	a_1	a_2	a_3
θ_1	3	1	0
θ_2	1	2	4

- (a) Plot the three risk points, describe $S \subseteq \mathbb{R}^2$, identify the lower boundary $\lambda(S)$.
- (b) Which nonrandomized actions are admissible?
- (c) Find the minimax risk point s_M by shrinking the lower quadrant $Q_\alpha = \{z : z_i \leq \alpha\}$. State V , write s_M as a convex combination of risk points, and give the minimax randomized strategy δ_M^* .
- (d) Identify the maximin prior π_M and verify $L(\theta, \delta_M^*) \leq V \leq L(\pi_M, a)$ for all θ, a .

The risk set

Read each column of the loss matrix as a point in $\mathbb{R}^2$ (coordinates indexed by θ_1, θ_2):

$$R(a_1) = (3, 1), \quad R(a_2) = (1, 2), \quad R(a_3) = (0, 4).$$

The risk set S is the convex hull, a closed triangle.

These three points are not collinear. The line through $(3, 1)$ and $(0, 4)$ is $x + y = 4$; the point $(1, 2)$ has $1 + 2 = 3 \neq 4$, so it lies strictly below. The triangle has nonempty interior. (This non-degeneracy is what makes the problem interesting: a collinear configuration would give a trivial single-segment lower boundary.)

Finding the lower boundary $\lambda(S)$

$\lambda(S)$ is the part of the boundary facing the lower-left: an edge is on $\lambda(S)$ iff its **outward** normal has both components nonpositive. (Then moving in any negative-orthant direction leaves S , which is the geometric meaning of “lower boundary.”)

Compute outward normals, using that the centroid $(4/3, 7/3)$ is interior.

- ▶ **Edge** $(3, 1) \rightarrow (1, 2)$: outward normal $\propto (-1, -2)$, both negative. **On** $\lambda(S)$.
- ▶ **Edge** $(1, 2) \rightarrow (0, 4)$: outward normal $\propto (-2, -1)$, both negative. **On** $\lambda(S)$.
- ▶ **Edge** $(3, 1) \rightarrow (0, 4)$: outward normal $\propto (1, 1)$, both positive. **Not on** $\lambda(S)$.

So $\lambda(S)$ is the two lower-left edges meeting at $(1, 2)$.

Admissibility

By the lower-boundary characterization of admissibility (Lecture 7, Theorem 8): a nonrandomized action is admissible iff its risk point lies on $\lambda(S)$.

All three risk points $(3, 1)$, $(1, 2)$, $(0, 4)$ are vertices of $\lambda(S)$, so **all three actions** a_1, a_2, a_3 **are admissible**.

Quick sanity check by direct domination: no risk point is coordinatewise $\leq$ another, so no action dominates another. Consistent with all three being admissible.

Teaching point. Admissibility here is purely geometric. There is no need to compute any risk integral; you just look at which points sit on the lower-left frontier.

The minimax point

Shrink the lower quadrant $Q_\alpha = \{z : z_1 \leq \alpha, z_2 \leq \alpha\}$ by decreasing α . The last value of α for which Q_α still touches S is the minimax value, and the touch point is s_M . Because Q_α has its corner on the diagonal $\{z_1 = z_2\}$, the touch point is where the diagonal meets $\lambda(S)$.

Parametrize edge $(3, 1) \rightarrow (1, 2)$ as $s(\tau) = (3 - 2\tau, 1 + \tau)$, $\tau \in [0, 1]$. Set $s_1 = s_2$:

$$3 - 2\tau = 1 + \tau \implies \tau = 2/3 \implies s_M = (5/3, 5/3).$$

(The other edge gives $\tau = -1/3 \notin [0, 1]$, no intersection.) So

$V = 5/3, \quad s_M = (5/3, 5/3).$

The minimax rule

Express s_M as a convex combination of the risk points on its edge:

$$s_M = (5/3, 5/3) = \frac{1}{3}(3, 1) + \frac{2}{3}(1, 2) = \frac{1}{3}R(a_1) + \frac{2}{3}R(a_2).$$

So the minimax randomized strategy plays a_1 with probability $1/3$ and a_2 with probability $2/3$:

$$\delta_M^* = \frac{1}{3}a_1 + \frac{2}{3}a_2.$$

Teaching point. The minimax rule is *randomized*. No single action is minimax here: a_1 has worst-case risk 3, a_2 has 2, a_3 has 4. Mixing a_1 and a_2 beats all three. This is the whole reason randomization matters in finite games.

The maximin prior and verification

The supporting line to S at s_M is edge E_1 , whose inward normal is $(1, 2)$ (both components nonnegative). Normalize to a probability vector:

$$\pi_M = \left(\frac{1}{3}, \frac{2}{3}\right).$$

Verification. Risk of δ_M^* at each θ :

$$L(\theta_1, \delta_M^*) = \frac{1}{3}(3) + \frac{2}{3}(1) = \frac{5}{3}, \quad L(\theta_2, \delta_M^*) = \frac{1}{3}(1) + \frac{2}{3}(2) = \frac{5}{3}.$$

Loss against π_M for each action:

$$L(\pi_M, a_1) = \frac{5}{3}, \quad L(\pi_M, a_2) = \frac{5}{3}, \quad L(\pi_M, a_3) = \frac{8}{3}.$$

So $L(\theta, \delta_M^*) = 5/3$ for every θ , and $L(\pi_M, a) \geq 5/3$ for every a . Condition (*) of the minimax theorem (Lecture 6, Theorem 2) holds: $V = 5/3$, δ_M^* is minimax, π_M is maximin.

Problem 2: Binomial minimax (equalizer-Bayes)

The question

$X \sim \text{Binomial}(n, \theta)$, $\theta \in [0, 1]$, $\mathcal{A} = [0, 1]$, squared-error loss.

- (a) For $\pi = \text{Beta}(\alpha, \beta)$, derive the Bayes rule $\delta_{\alpha, \beta}(X)$.
- (b) Compute $R(\theta, \delta_{\alpha, \beta})$ as a polynomial in θ over $(\alpha + \beta + n)^2$.
- (c) Find the unique (α^*, β^*) making the risk constant in θ . Give V and the equalizer rule.
- (d) Conclude δ^* is minimax. Identify the maximin prior.

Bayes rule and risk

(a) Beta-Binomial conjugacy: posterior is $\text{Beta}(\alpha + X, \beta + n - X)$. Under squared error, the Bayes rule is the posterior mean:

$$\delta_{\alpha,\beta}(X) = \frac{\alpha + X}{\alpha + \beta + n}.$$

(b) Write $m = \alpha + \beta + n$. Using $\mathbb{E}_\theta[X] = n\theta$ and $\text{Var}_\theta(X) = n\theta(1 - \theta)$:

$$\text{bias} = \frac{\alpha - (\alpha + \beta)\theta}{m}, \quad \text{Var}_\theta(\delta_{\alpha,\beta}) = \frac{n\theta(1 - \theta)}{m^2}.$$

Combining and expanding,

$$R(\theta, \delta_{\alpha,\beta}) = \frac{[(\alpha + \beta)^2 - n]\theta^2 + [n - 2\alpha(\alpha + \beta)]\theta + \alpha^2}{(\alpha + \beta + n)^2}.$$

The equalizer condition

(c) For the risk to be constant in θ , the coefficients of θ^2 and θ in the numerator must both vanish:

$$(\alpha + \beta)^2 - n = 0, \quad n - 2\alpha(\alpha + \beta) = 0.$$

From the first, $\alpha + \beta = \sqrt{n}$. Substituting into the second, $n - 2\alpha\sqrt{n} = 0$, so $\alpha = \sqrt{n}/2$, and $\beta = \sqrt{n} - \alpha = \sqrt{n}/2$:

$$\alpha^* = \beta^* = \frac{\sqrt{n}}{2}.$$

The constant risk is $\alpha^{*2}/m^{*2} = (n/4)/(\sqrt{n} + n)^2$. Factor $\sqrt{n} + n = \sqrt{n}(1 + \sqrt{n})$:

$$V = \frac{n/4}{n(1 + \sqrt{n})^2} = \frac{1}{4(1 + \sqrt{n})^2}, \quad \delta^*(X) = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}.$$

Why constant risk plus Bayes equals minimax

(d) Set $\pi_M = \text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ and $\delta^* = \delta_{\alpha^*, \beta^*}$. Two facts in hand:

- ▶ (i) **Constant risk:** $R(\theta, \delta^*) = V$ for all θ (from (c)).
- ▶ (ii) **Bayes property:** δ^* is Bayes against π_M (from (a)).

For any rule δ :

$$\sup_{\theta} R(\theta, \delta^*) \stackrel{(i)}{=} V = \int_0^1 V \pi_M(d\theta) \stackrel{(i)}{=} \int_0^1 R(\theta, \delta^*) \pi_M(d\theta) = r(\pi_M, \delta^*) \stackrel{(ii)}{\leq} r(\pi_M, \delta) \leq \sup_{\theta} R(\theta, \delta).$$

The last step uses that Bayes risk (an average) never exceeds the sup risk. So $\sup_{\theta} R(\theta, \delta^*) \leq \sup_{\theta} R(\theta, \delta)$ for every δ : δ^* is minimax, π_M is maximin.

Teaching point. This is the equalizer-Bayes route (Lecture 8, Example 17). The hard work is *finding* the prior whose Bayes rule has constant risk. Once you have it, minimaxity is two lines.

A surprise worth noticing

The minimax estimator $\delta^*(X) = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}$ is **not** the sample proportion X/n .

It shrinks toward $1/2$. Rewrite it as a weighted average:

$$\delta^*(X) = \frac{n}{n + \sqrt{n}} \cdot \frac{X}{n} + \frac{\sqrt{n}}{n + \sqrt{n}} \cdot \frac{1}{2}.$$

The weight on $1/2$ is $\sqrt{n}/(n + \sqrt{n}) \rightarrow 0$ as $n \rightarrow \infty$, so for large n the minimax rule is close to X/n . But for finite n , the minimax rule is biased on purpose: the worst case for X/n is at $\theta = 1/2$, where its variance is largest, and the shrinkage trades a little bias for reduced worst-case risk.

Problem 3: Minimax via a sequence of priors

The question

$X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$, squared-error loss, $\delta_0(X) = X$.

- (a) Compute $R(\theta, \delta_0)$ and $\sup_{\theta} R(\theta, \delta_0)$.
- (b) For $\pi_k = N(0, k)$, derive the Bayes rule and minimum Bayes risk.
- (c) Apply the sequence-of-priors theorem to conclude δ_0 is minimax with $V = 1$. Explain why no single proper prior suffices.

Risk and Bayes rule

(a) $\delta_0(X) = X$ is unbiased with $\text{Var}_\theta(X) = 1$:

$$R(\theta, \delta_0) = 1 \text{ for all } \theta, \quad \sup_{\theta} R(\theta, \delta_0) = 1.$$

(b) Normal-Normal conjugacy. The posterior of θ given X is $N(\mu_k, \tau_k^2)$ where the precisions add:

$$\frac{1}{\tau_k^2} = \frac{1}{k} + 1 \implies \tau_k^2 = \frac{k}{k+1}, \quad \mu_k = \frac{k}{k+1}X.$$

The Bayes rule is the posterior mean $\delta_{\pi_k}(X) = \frac{k}{k+1}X$, and the minimum Bayes risk equals the (constant) posterior variance:

$$r(\pi_k, \delta_{\pi_k}) = \tau_k^2 = \frac{k}{k+1}.$$

The limiting argument

(c) Apply Theorem 3 of Lecture 6 with the sequence $\{\pi_k\}$ and candidate δ_0 . The hypothesis reduces to

$$\sup_{\theta} R(\theta, \delta_0) \leq \liminf_{k \rightarrow \infty} r(\pi_k, \delta_{\pi_k}).$$

LHS = 1 from (a). RHS: $r(\pi_k, \delta_{\pi_k}) = k/(k+1) \rightarrow 1$, so $\liminf = 1$. Hence $1 \leq 1$, with equality. Theorem 3 gives: $\delta_0(X) = X$ is minimax, $V = 1$.

Why no single proper prior works. For any proper prior π with finite variance, $r(\pi, \delta_{\pi}) < 1 = \sup_{\theta} R(\theta, \delta_0)$. (For $\pi = N(0, \tau^2)$, the Bayes risk is $\tau^2/(1 + \tau^2) < 1$.) So the equalizer-Bayes argument of Problem 2 cannot fire with any single prior: the sandwich would need $r(\pi, \delta_0) = 1$, but no proper prior achieves it. The sequence $\pi_k = N(0, k)$ approximates the improper flat prior, the natural least-favorable “prior” here, which is not a usable prior on its own. The sequence makes the limiting argument rigorous.

Problems 2 and 3 side by side

The two problems are the two faces of minimax verification.

- ▶ **Problem 2 (proper prior exists).** The least-favorable prior is a genuine $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$. The equalizer rule is Bayes against it, with Bayes risk exactly equal to its constant risk. One prior does the whole job.
- ▶ **Problem 3 (no proper prior works).** The least-favorable “prior” is the improper flat prior on $\mathbb{R}$. No proper prior gives Bayes risk 1, so we approximate with a sequence whose Bayes risks climb to 1.

The difference traces back to compactness: $[0, 1]$ is compact, so the binomial problem has a genuine least-favorable prior; $\mathbb{R}$ is not, so the Normal-mean problem needs a limiting sequence.

Problem 4: Restricted-parameter minimax

The question

$X \sim N(\theta, 1)$, now with $\theta \in [-m, m]$, squared-error loss.

(a) For the two-point prior $\pi_m = \frac{1}{2}\delta_{-m} + \frac{1}{2}\delta_m$, derive the Bayes rule and show $\delta_{\pi_m}(X) = m \tanh(mX)$.

(b) Show $R(m, \delta_{\pi_m}) = R(-m, \delta_{\pi_m})$ by symmetry.

(c) Assuming $\sup_{\theta \in [-m, m]} R(\theta, \delta_{\pi_m}) = R(m, \delta_{\pi_m})$, verify δ_{π_m} is minimax. State V .

(d) Find the pointwise limit of $m \tanh(mX)$ as $m \rightarrow \infty$ and compare with Problem 3.

The two-point Bayes rule

(a) The posterior is supported on $\{-m, m\}$ with

$$\mathbb{P}(\theta = m \mid X) = \frac{\phi(X - m)}{\phi(X - m) + \phi(X + m)}.$$

Multiply numerator and denominator by $e^{(X^2+m^2)/2}$. Since $\phi(X - m)e^{(X^2+m^2)/2} = e^{mX}$ and $\phi(X + m)e^{(X^2+m^2)/2} = e^{-mX}$,

$$\mathbb{P}(\theta = m \mid X) = \frac{e^{mX}}{e^{mX} + e^{-mX}}, \quad \mathbb{P}(\theta = -m \mid X) = \frac{e^{-mX}}{e^{mX} + e^{-mX}}.$$

The Bayes rule is the posterior mean:

$$\delta_{\pi_m}(X) = m \cdot \frac{e^{mX} - e^{-mX}}{e^{mX} + e^{-mX}} = m \tanh(mX).$$

Risk symmetry and minimaxity

(b) $R(\theta, \delta_{\pi_m}) = \mathbb{E}_\theta[(m \tanh(mX) - \theta)^2]$. At $\theta = -m$, substitute $Y = -X$ (so $Y \sim N(m, 1)$) and use that $\tanh$ is odd:

$$R(-m, \delta_{\pi_m}) = \mathbb{E}_{Y \sim N(m, 1)}[(m - m \tanh(mY))^2] = R(m, \delta_{\pi_m}).$$

(c) Since π_m puts mass $1/2$ on each of $\pm m$, the symmetry gives $r(\pi_m, \delta_{\pi_m}) = R(m, \delta_{\pi_m})$. Now the standard sandwich, for any δ :

$$\sup_{\theta} R(\theta, \delta_{\pi_m}) = R(m, \delta_{\pi_m}) = r(\pi_m, \delta_{\pi_m}) \leq r(\pi_m, \delta) \leq \sup_{\theta} R(\theta, \delta).$$

The first equality is the assumption in (c); the middle is the Bayes optimality of δ_{π_m} . So δ_{π_m} is minimax over $[-m, m]$ with $V = R(m, \delta_{\pi_m}) = r(\pi_m, \delta_{\pi_m})$.

The limit, and why it is striking

(d) For fixed X and $m \rightarrow \infty$:

$$m \tanh(mX) \rightarrow \begin{cases} +\infty & X > 0, \\ 0 & X = 0, \\ -\infty & X < 0. \end{cases}$$

So $m \tanh(mX) \not\rightarrow X$. The restricted minimax rule does **not** recover the unrestricted minimax rule $\delta_0(X) = X$ in the limit.

Teaching point. Near $X = 0$, $m \tanh(mX) \approx m^2 X$, which overshoots X by a factor of m^2 . The restricted minimax rule exploits the constraint $\theta \in [-m, m]$ aggressively, shrinking toward the support $\{-m, m\}$ of the least-favorable prior. Relaxing the constraint ($m \rightarrow \infty$) does not smoothly hand back the unrestricted solution: the rule blows up instead. The constraint set changes the minimax problem qualitatively, not just quantitatively.

Wrap-up

The four routes to minimax

Problem	Setting	Route	Least-favorable prior
1	Finite game	Risk-set geometry	$\pi_M = (1/3, 2/3)$
2	Binomial $[0, 1]$	Equalizer-Bayes	Beta($\sqrt{n}/2, \sqrt{n}/2$)
3	Normal mean $\mathbb{R}$	Sequence of priors	Flat prior (improper, approximated)
4	Normal mean $[-m, m]$	Two-point prior	$\frac{1}{2}\delta_{-m} + \frac{1}{2}\delta_m$

The single unifying idea: **a Bayes rule whose Bayes risk equals its worst-case risk is minimax**. When the parameter space is compact (Problems 1, 2, 4), a genuine least-favorable prior exists and the argument is one sandwich. When it is not (Problem 3), the prior becomes a limit and the argument becomes a limiting inequality.

The recurring lesson across Problems 2 and 4: **minimax rules are biased**. They sacrifice performance at typical parameter values to protect against the worst case. The binomial rule shrinks toward $1/2$; the restricted-mean rule shrinks toward $\{-m, m\}$. Bias is not a defect of these rules, it is the mechanism by which they control worst-case risk.